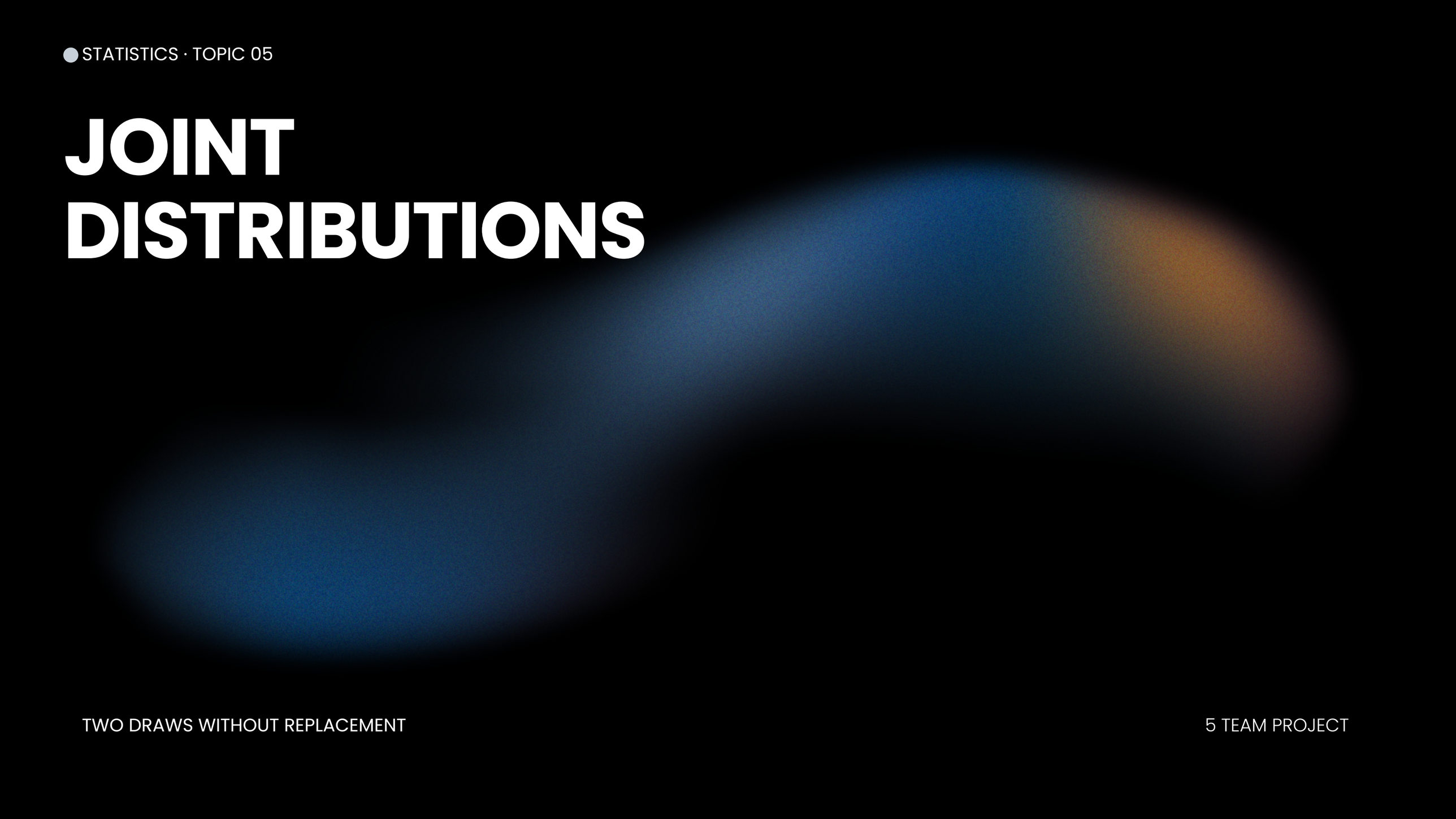
STATISTICS · TOPIC 05

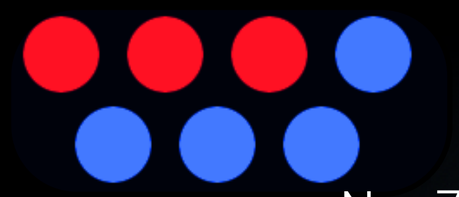
JOINT DISTRIBUTIONS



TWO DRAWS WITHOUT REPLACEMENT

5 TEAM PROJECT

THE JOINT MODEL



N = 7, K = 3

N = total number of balls
K = number of red balls
N - K = number of non-red balls

X_1 = 1 if the first ball is red; 0 otherwise
X_2 = 1 if the second ball is red; 0 otherwise

Two consecutive draws without replacement.

JOINT PROBABILITY DISTRIBUTION

P(1,1) = K(K-1) / [N(N-1)]

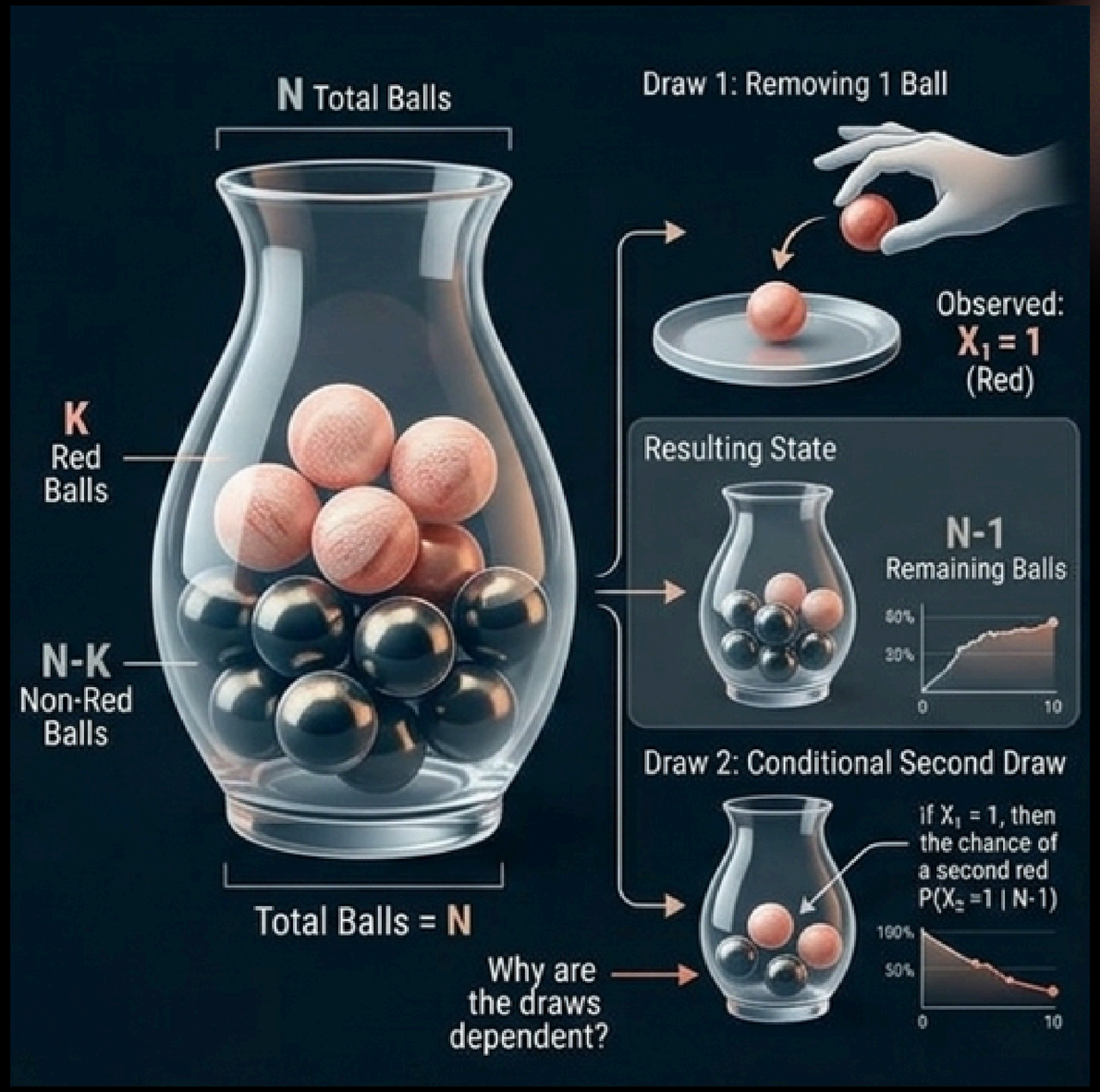
P(1,0) = K(N-K) / [N(N-1)]

P(0,1) = K(N-K) / [N(N-1)]

P(0,0) = (N-K)(N-K-1) / [N(N-1)]

VALIDITY: $\sum^{1}\sum^{1} P(X_1 = i, X_2 = j) = 1$

$$(X_1, X_2) \in \{(0,0), (0,1), (1,0), (1,1)\}$$



MARGINAL & CONDITIONAL DISTRIBUTIONS

$P(X_1=1) = P(X_2=1) = K/N$

Marginal distributions:
$P(X_1 = 1) = P(X_2 = 1) = K / N$
$P(X_1 = 0) = P(X_2 = 0) = (N - K) / N$

Conditional distributions Given $X_1 = 1$: $P(X_2 = 1 \mid X_1 = 1) = (K - 1) / (N - 1)$
Given $X_1 = 0$: $P(X_2 = 1 \mid X_1 = 0) = K / (N - 1)$

Why are the draws dependent? $(K - 1) / (N - 1) \neq K / N$ Removing the first ball changes the composition of the urn.

MOMENTS: EXPECTATION & VARIANCE

DEPENDENCE: COVARIANCE & CORRELATION

01 Expectation

$$E(X_1) = E(X_2) = K/N = p$$

The second ball is red with the same probability as the first.

02 Variance

$$Var(X_i) = pq$$

$$q = 1 - p$$

Each X_i is 0 or 1 (Bernoulli), so Var = $p - p^2 = pq$.

03 Covariance

$$Cov(X_1, X_2) = -K(N-K) / [N^2(N-1)]$$

$= -pq/(N-1) < 0$

04 Correlation

$$\rho = -1/(N-1)$$

Does not depend on K. $\rho \to 0$ when N is large.

Part B · N balls, K red · $p = K/N$ · $X_i = 1$ if draw i is red

DEPENDENT, YET EXCHANGEABLE

1 WHY ARE THEY DEPENDENT?

Independence would require the joint probability to equal the product of the marginals.

Joint probability

$$\mathbf{P(X_1 = 1, X_2 = 1) = K(K - 1) / N(N - 1)}$$

Product of marginals

$$\mathbf{P(X_1 = 1)P(X_2 = 1) = (K/N)^2}$$

$$\neq$$

THEREFORE, X_1 AND X_2 ARE **DEPENDENT**.

2 WHY ARE THEY EXCHANGEABLE?

Exchangeability means that swapping the draw labels leaves the joint distribution unchanged.

The two mixed outcomes have equal probability:

$$\mathbf{P(X_1 = 1, X_2 = 0) = P(X_1 = 0, X_2 = 1)}$$

Both equal $K(N - K) / [N(N - 1)]$.

Exchangeable

DEPENDENCE DOES NOT PREVENT EXCHANGEABILITY

THE SUM: DISTRIBUTION, MEAN & VARIANCE

3 HYPERGEOMETRIC DISTRIBUTION

$S = X_1 + X_2$ the number of red balls in two draws

For s = 0, 1, 2:

$$P(S=s)=C(K,s)C(N-K,2-s)/C(N,2)$$

s	P(S = s)
0	(N − K)(N − K − 1) / [N(N − 1)]
1	2K(N − K) / [N(N − 1)]
2	K(K − 1) / [N(N − 1)]

Hence S ~ Hypergeometric(N, K, 2)

4 MEAN AND VARIANCE

Expected value

$$E(S) = 0P_0 + P_1 + 2P_2 = 2K/N = 2p$$

Use the covariance identity:

$Var(S)=Var(X_1)+Var(X_2)+2Cov(X_1,X_2)=$

$=2pq - 2pq/(N - 1)$

$$Var(S) = 2pq\,(N - 2) / (N - 1)$$

Hypergeometric check

$Var(S) = npq\,(N - n)/(N - 1)$, $n = 2$

$= 2pq\,(N - 2)/(N - 1)$

TRANSFORMATIONS AND SPECIAL RESULTS

n draws without replacement • N balls • K red

1 Indicator setup

$$S_n = I_1 + I_2 + \cdots + I_n$$

$I_i = 1$ if draw i is red; otherwise 0

$$p = K/N \quad q = 1 - p$$

$$\mathrm{Var}(I_i) = pq$$

For distinct draws i and j:

$$P(\text{both red}) = K(K-1) / [N(N-1)]$$

$$\mathrm{Cov}(I_i, I_j) = -pq/(N-1)$$

Negative: a red ball is removed.

2 Count every covariance pair

There are $C(n, 2) = n(n-1)/2$ distinct pairs.

$$\mathrm{Var}(S_n) = \Sigma\, \mathrm{Var}(I_i) + 2\, \Sigma\, \mathrm{Cov}(I_i, I_j)$$

$$= npq + 2 \cdot [n(n-1)/2] \cdot [-pq/(N-1)]$$

$$= npq\,[1 - (n-1)/(N-1)]$$

$$\mathrm{Var}(S_n) = npq\,(N-n)/(N-1)$$

FPC for variance = (N−n)/(N−1)

For standard deviation: take the square root.

3 Large-population limit

Fixed n; N → ∞

$K/N \to p, \quad q = 1 - p$

Hypergeometric(N,K,n)

→ Binomial(n,p)

$$\mathrm{Var}(S_n) \to np(1-p)$$

Removal becomes negligible; draws become asymptotically independent.

Check: n = N ⇒ $\mathrm{Var}(S_n) = 0$

WITH replacement: Var = npq | **WITHOUT replacement: Var = npq × FPC**

Worked Example

Problem. Inspecting 2 of 10 Chargers. N = 10 items, K = 3 defective, 2 drawn without replacement. $X_i = 1$ if draw i is defective; $S = X_1 + X_2$.

Joint pmf. By the multiplication rule,
$p(x_1,x_2) = P(X_1=x_1)\cdot P(X_2=x_2 \mid X_1=x_1)$:

	$X_2=0$	$X_2=1$	Total
$X_1=0$	42/90	21/90	63/90
$X_1=1$	21/90	6/90	27/90
Total	63/90	27/90	1

Check: 42+21+21+6 = 90, so the pmf is valid.

Joint:
$P(X_1=1, X_2=1) = (3/10)(2/9) = 1/15 \approx 0.067$

Conditional:
$P(X_2=1 \mid X_1=1) = 2/9 \approx 0.222$ (vs. 0.3 unconditionally)

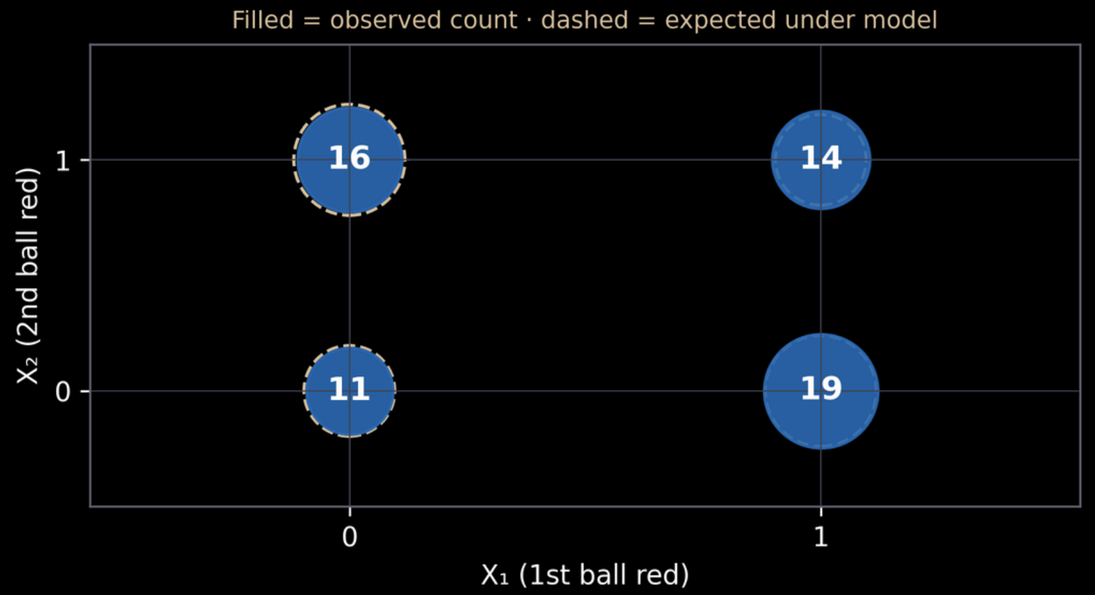
Sum:
$P(S=0, 1, 2) = 7/15, 7/15, 1/15$; $P(S \geq 1) = 8/15 \approx 0.533$

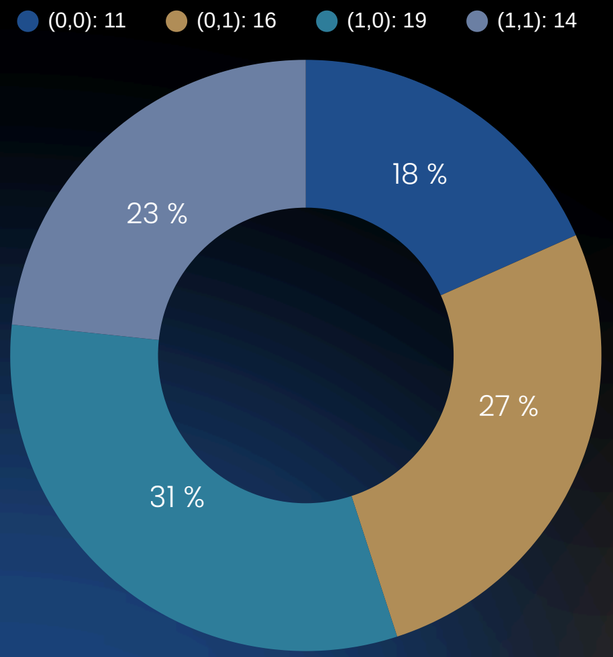
DATA EXAMPLE

60 simulated pairs (X_1, X_2): urn with N = 6 balls, K = 3 red, two draws without replacement

Data are simulated (numpy, seed 2026); no real data set used. Model: ρ = −1/(N−1) = −0.20.

Quantity	Sample (n = 60)	Theory
Mean X_1	0.550	0.500
Mean X_2	0.500	0.500
Variance s^2 of X_1	0.252	0.250
Variance s^2 of X_2	0.254	0.250
Covariance sxy	**−0.042**	**−0.050**
Correlation r	**−0.168**	**−0.200**





Joint frequency of (X_1, X_2)
Expected (n·P): 12 · 18 · 18 · 12
Theory P: 20% · 30% · 30% · 20%
P(X_2=1 | X_1=1): 0.424 vs 0.400
P(X_2=1 | X_1=0): 0.593 vs 0.600

INTERESTING FACTS

HISTORY

Fisher, 1935: lady tasting tea

- Muriel Bristol claimed she could tell milk-first from tea-first: 8 cups, 4 of each, random order.
- C(8,4) = 70 choices, so all correct by chance: 1/70 ≈ 0.014; at least 3 of 4: 17/70 ≈ 0.243.
- Correct picks ~ Hypergeometric(N=8, K=4, n=4), our model with n = 4.

HISTORY

de Finetti, 1931: exchangeable

- Our pair is exchangeable, $P(x_1,x_2) = P(x_2,x_1)$, yet $p(1,1) < p^2$: not independent.
- Urn draws without replacement are the standard exchangeable-but-dependent example.
- De Finetti's theorem (1931) is for infinite sequences; urns are only finite.

APPLICATION

Survey sampling: FPC

- Sampling without replacement shrinks the standard error by $\sqrt{((N-n)/(N-1))}$: our Part D result.
- n = 100 of N = 500: $\sqrt{(400/499)} = 0.895$, so the standard error is about 10% smaller.
- Below 5% sampled the factor is ≈ 1 and ignored.

LINK

Topics 1 and 3: the lowest possible correlation

- Indicators give a 2×2 table (Topic 1); $\rho = -1/(N-1)$ is its phi coefficient.
- Floor: $0 \le \mathrm{Var}(X_1+\ldots+X_N) = N\sigma^2[1 + (N-1)\rho]$, so $\rho \ge -1/(N-1)$.
- Check: N = 2, K = 1 gives $\rho = -1$; N = 52 gives −0.020; $N \to \infty$ gives $\rho \to 0$ and $S \to \mathrm{Bin}(2, p)$ (Topic 3).

MISCONCEPTION

“The second ball has a different chance”

- False before draw 1 is seen: $P(X_2=1) = (K/N)(K-1)/(N-1) + ((N-K)/N)\cdot K/(N-1) = K/N$.
- Deck: the 2nd card is an ace with probability 4/52 = 1/13, like the 1st: (12 + 192)/2652 = 1/13.
- Only the conditional chance changes: 3/51 after an ace, 4/51 after a non-ace.